Supplementary Material: Individual RL Expert Performance

Table S1. Return performance of the individual experts across five seeds.

Expert	AR (%) Mean ± SD	AR (%) Min / Max	CR (%) Mean ± SD	CR (%) Min / Max
S&P 500				
DDPG	17.99 ± 1.81	15.77 / 19.49	420.54 ± 76.83	327.49 / 484.81
TD3	16.99 ± 2.17	14.91 / 20.09	380.08 ± 90.68	297.03 / 514.67
SAC	15.60 ± 0.03	15.56 / 15.63	321.33 ± 0.98	319.98 / 322.45
PPO	16.60 ± 0.60	15.68 / 17.17	359.34 ± 23.07	324.27 / 381.78
A2C	17.93 ± 2.23	15.82 / 21.07	420.21 ± 100.42	329.30 / 566.22
NSE				
DDPG	21.19 ± 0.39	20.56 / 21.63	460.72 ± 16.02	434.93 / 479.03
TD3	21.14 ± 1.41	18.88 / 22.44	460.69 ± 56.70	371.65 / 514.36
SAC	21.65 ± 0.04	21.63 / 21.71	479.91 ± 1.54	478.81 / 482.54
PPO	21.90 ± 0.49	21.44 / 22.66	491.08 ± 21.72	470.77 / 524.55
A2C	21.56 ± 1.87	18.98 / 24.21	479.69 ± 80.10	375.26 / 598.95
Taiwan				
DDPG	18.41 ± 3.44	12.61 / 21.27	420.05 ± 127.12	212.95 / 537.38
TD3	15.31 ± 4.71	8.11 / 19.40	313.21 ± 140.90	111.45 / 449.06
SAC	18.96 ± 0.03	18.93 / 18.98	429.59 ± 1.16	428.32 / 430.80
PPO	18.43 ± 1.13	16.80 / 19.73	409.18 ± 45.98	344.19 / 463.80
A2C	17.30 ± 3.21	12.72 / 20.16	374.00 ± 115.70	215.78 / 483.33
DSE				
DDPG	1.45 ± 1.29	0.07 / 3.20	13.64 ± 12.43	0.59 / 31.04
TD3	2.34 ± 2.05	-1.19 / 3.74	23.12 ± 19.37	-9.78 / 37.01
SAC	2.15 ± 0.06	2.08 / 2.23	19.99 ± 0.62	19.31 / 20.86
PPO	2.01 ± 0.64	1.29 / 2.91	18.74 ± 6.45	11.59 / 27.87
A2C	0.09 ± 1.77	-1.62 / 2.36	1.58 ± 15.64	-13.04 / 22.13

Notes: Each expert is evaluated using seeds 42, 123, 456, 789 and 2026. SD is the sample standard deviation across the five seeds (denominator n - 1); Min and Max are the observed seed extrema. AR is annualized compounded return; CR is cumulative return. Percentage-valued metrics are shown in percent, and their SDs are in percentage points.

These are descriptive summaries of training-seed variability over a common historical evaluation period within each market; they are not confidence intervals or tests of statistical superiority.

Table S2. Risk-adjusted performance and maximum drawdown across five seeds.

Expert	SR Mean ± SD	SoR Mean ± SD	MDD (%) Mean ± SD
S&P 500			
DDPG	0.84 ± 0.06	1.19 ± 0.08	36.77 ± 0.74
TD3	0.81 ± 0.07	1.14 ± 0.10	36.72 ± 0.75
SAC	0.78 ± 0.00	1.10 ± 0.00	37.11 ± 0.02
PPO	0.81 ± 0.02	1.14 ± 0.03	36.72 ± 0.35
A2C	0.83 ± 0.07	1.18 ± 0.11	36.69 ± 1.07
NSE			
DDPG	1.04 ± 0.01	1.38 ± 0.02	59.24 ± 0.85
TD3	1.06 ± 0.06	1.40 ± 0.08	56.71 ± 1.83
SAC	1.09 ± 0.00	1.44 ± 0.00	57.57 ± 0.06
PPO	1.10 ± 0.02	1.46 ± 0.03	57.00 ± 0.95
A2C	1.06 ± 0.08	1.42 ± 0.13	56.19 ± 4.89
Taiwan			
DDPG	1.12 ± 0.09	1.53 ± 0.12	30.11 ± 3.26
TD3	1.00 ± 0.16	1.37 ± 0.21	27.67 ± 4.39
SAC	1.18 ± 0.00	1.59 ± 0.00	30.08 ± 0.03
PPO	1.14 ± 0.06	1.55 ± 0.08	30.15 ± 0.56
A2C	1.06 ± 0.20	1.44 ± 0.28	29.73 ± 2.45
DSE			
DDPG	0.17 ± 0.07	0.26 ± 0.10	47.96 ± 2.40
TD3	0.22 ± 0.09	0.34 ± 0.15	53.17 ± 3.02
SAC	0.21 ± 0.00	0.32 ± 0.01	45.45 ± 0.19
PPO	0.20 ± 0.04	0.30 ± 0.05	48.52 ± 1.75
A2C	0.10 ± 0.09	0.15 ± 0.14	50.67 ± 1.92

Notes: SR denotes the annualized Sharpe ratio; SoR denotes the annualized Sortino ratio. MDD is the maximum drawdown, reported as a positive loss magnitude; lower values indicate smaller peak-to-trough losses. Means and sample SDs are computed across five seeds. Minimum and maximum values for all metrics are included in the accompanying summary CSV.

Metric definitions: AR = 100 × [(product of (1 + daily return)) raised to (252 / T) - 1]; CR = 100 × (terminal portfolio value - 1), with initial portfolio value normalized to 1. SR uses daily excess returns and their sample SD, annualized by the square root of 252. SoR uses daily excess returns and the root mean squared negative excess return, annualized by the square root of 252. Saved evaluation metrics are reported without changing their original conventions.

Table S3. Individual expert performance by seed: S&P 500.

Out-of-sample period: 2016-01-06 to 2025-12-12; 2,500 daily observations per expert and seed.

Expert	Seed	AR (%)	CR (%)	SR	SoR	MDD (%)
DDPG	42	16.29	346.94	0.78	1.10	37.62
DDPG	123	19.49	484.81	0.88	1.24	37.24
DDPG	456	19.02	462.56	0.87	1.23	36.90
DDPG	789	15.77	327.49	0.77	1.09	35.74
DDPG	2026	19.40	480.90	0.90	1.27	36.33
TD3	42	16.73	364.04	0.81	1.14	36.65
TD3	123	14.91	297.03	0.72	1.01	37.73
TD3	456	15.09	303.21	0.75	1.06	36.98
TD3	789	18.11	421.46	0.86	1.21	35.67
TD3	2026	20.09	514.67	0.89	1.26	36.57
SAC	42	15.62	321.84	0.78	1.10	37.10
SAC	123	15.56	319.98	0.78	1.10	37.10
SAC	456	15.58	320.71	0.78	1.10	37.14
SAC	789	15.63	322.45	0.78	1.10	37.10
SAC	2026	15.61	321.66	0.78	1.10	37.09
PPO	42	16.92	371.60	0.82	1.15	36.76
PPO	123	16.33	348.44	0.80	1.13	36.61
PPO	456	15.68	324.27	0.78	1.09	36.98
PPO	789	16.90	370.59	0.82	1.16	36.18
PPO	2026	17.17	381.78	0.82	1.16	37.07
A2C	42	17.04	376.41	0.82	1.15	36.34
A2C	123	15.82	329.30	0.78	1.10	36.62
A2C	456	16.33	348.34	0.76	1.07	38.44
A2C	789	19.40	480.79	0.89	1.26	36.55
A2C	2026	21.07	566.22	0.93	1.32	35.51

Notes: AR = annualized compounded return; CR = cumulative return; SR = Sharpe ratio; SoR = Sortino ratio; MDD = maximum drawdown. AR, CR and MDD are expressed in percent. Values are rounded to two decimal places; the accompanying CSV retains higher precision. Each row is one standalone expert evaluation.

Table S4. Individual expert performance by seed: NSE.

Out-of-sample period: 2016-01-05 to 2025-02-25; 2,260 daily observations per expert and seed.

Expert	Seed	AR (%)	CR (%)	SR	SoR	MDD (%)
DDPG	42	21.31	465.55	1.05	1.39	58.04
DDPG	123	21.23	462.06	1.04	1.38	60.04
DDPG	456	21.23	462.01	1.05	1.39	58.68
DDPG	789	20.56	434.93	1.03	1.36	59.56
DDPG	2026	21.63	479.03	1.03	1.36	59.86
TD3	42	21.01	453.09	1.06	1.40	53.97
TD3	123	21.15	459.02	1.03	1.36	57.65
TD3	456	22.44	514.36	1.09	1.45	56.99
TD3	789	18.88	371.65	0.98	1.29	58.84
TD3	2026	22.23	505.32	1.12	1.49	56.08
SAC	42	21.63	478.81	1.09	1.44	57.51
SAC	123	21.65	480.01	1.09	1.44	57.59
SAC	456	21.64	479.27	1.09	1.44	57.60
SAC	789	21.71	482.54	1.10	1.45	57.51
SAC	2026	21.63	478.90	1.09	1.44	57.64
PPO	42	22.11	500.02	1.12	1.48	55.50
PPO	123	22.66	524.55	1.13	1.50	57.26
PPO	456	21.74	483.88	1.09	1.44	57.96
PPO	789	21.56	476.19	1.09	1.44	56.72
PPO	2026	21.44	470.77	1.08	1.42	57.57
A2C	42	21.35	467.23	1.10	1.45	53.75
A2C	123	24.21	598.95	1.17	1.60	49.15
A2C	456	21.94	492.28	1.07	1.42	58.81
A2C	789	21.29	464.70	1.04	1.39	57.42
A2C	2026	18.98	375.26	0.94	1.24	61.81

Notes: AR = annualized compounded return; CR = cumulative return; SR = Sharpe ratio; SoR = Sortino ratio; MDD = maximum drawdown. AR, CR and MDD are expressed in percent. Values are rounded to two decimal places; the accompanying CSV retains higher precision. Each row is one standalone expert evaluation.

Table S5. Individual expert performance by seed: Taiwan.

Out-of-sample period: 2016-01-06 to 2025-12-18; 2,420 daily observations per expert and seed.

Expert	Seed	AR (%)	CR (%)	SR	SoR	MDD (%)
DDPG	42	12.61	212.95	0.98	1.35	24.59
DDPG	123	19.45	450.99	1.13	1.54	32.46
DDPG	456	20.53	500.91	1.20	1.64	31.98
DDPG	789	18.20	398.06	1.09	1.48	29.75
DDPG	2026	21.27	537.38	1.20	1.64	31.78
TD3	42	17.64	376.06	1.09	1.49	29.93
TD3	123	18.37	405.01	1.14	1.55	30.05
TD3	456	19.40	449.06	1.11	1.51	31.36
TD3	789	8.11	111.45	0.78	1.06	20.51
TD3	2026	13.04	224.49	0.90	1.23	26.49
SAC	42	18.98	430.45	1.18	1.60	30.08
SAC	123	18.98	430.80	1.18	1.59	30.12
SAC	456	18.96	429.98	1.18	1.60	30.09
SAC	789	18.93	428.39	1.17	1.59	30.08
SAC	2026	18.93	428.32	1.18	1.59	30.03
PPO	42	17.86	384.42	1.11	1.51	29.61
PPO	123	16.80	344.19	1.06	1.44	30.38
PPO	456	19.73	463.80	1.21	1.64	29.56
PPO	789	18.97	430.03	1.17	1.59	30.31
PPO	2026	18.81	423.47	1.15	1.56	30.88
A2C	42	12.72	215.78	0.78	1.07	32.14
A2C	123	15.16	287.85	0.90	1.23	29.05
A2C	456	20.16	483.33	1.20	1.63	30.81
A2C	789	19.30	444.63	1.18	1.61	30.84
A2C	2026	19.16	438.39	1.22	1.68	25.80

Notes: AR = annualized compounded return; CR = cumulative return; SR = Sharpe ratio; SoR = Sortino ratio; MDD = maximum drawdown. AR, CR and MDD are expressed in percent. Values are rounded to two decimal places; the accompanying CSV retains higher precision. Each row is one standalone expert evaluation.

Table S6. Individual expert performance by seed: DSE.

Out-of-sample period: 2016-01-05 to 2025-03-04; 2,160 daily observations per expert and seed.

Expert	Seed	AR (%)	CR (%)	SR	SoR	MDD (%)
DDPG	42	1.97	18.21	0.20	0.30	49.74
DDPG	123	0.07	0.59	0.10	0.15	50.27
DDPG	456	3.20	31.04	0.27	0.41	44.37
DDPG	789	0.31	2.71	0.11	0.17	48.57
DDPG	2026	1.71	15.65	0.18	0.28	46.85
TD3	42	3.74	37.01	0.28	0.44	53.31
TD3	123	3.68	36.31	0.28	0.43	49.26
TD3	456	3.12	30.16	0.26	0.40	51.17
TD3	789	-1.19	-9.78	0.06	0.09	56.64
TD3	2026	2.34	21.92	0.22	0.33	55.45
SAC	42	2.12	19.70	0.21	0.31	45.32
SAC	123	2.23	20.86	0.21	0.32	45.52
SAC	456	2.19	20.36	0.21	0.32	45.35
SAC	789	2.08	19.31	0.21	0.31	45.76
SAC	2026	2.12	19.72	0.21	0.31	45.31
PPO	42	2.08	19.29	0.20	0.31	48.46
PPO	123	1.51	13.71	0.17	0.26	47.69
PPO	456	1.29	11.59	0.16	0.24	51.52
PPO	789	2.27	21.24	0.21	0.32	47.01
PPO	2026	2.91	27.87	0.25	0.38	47.93
A2C	42	2.36	22.13	0.22	0.33	50.51
A2C	123	-1.62	-13.04	0.01	0.02	52.86
A2C	456	-0.41	-3.47	0.07	0.11	51.23
A2C	789	-1.41	-11.44	0.01	0.01	51.17
A2C	2026	1.51	13.71	0.17	0.26	47.60

Notes: AR = annualized compounded return; CR = cumulative return; SR = Sharpe ratio; SoR = Sortino ratio; MDD = maximum drawdown. AR, CR and MDD are expressed in percent. Values are rounded to two decimal places; the accompanying CSV retains higher precision. Each row is one standalone expert evaluation.